TACT: Trust-Anchored Confidence Tempering for Self-Consistency Voting in Large Language Models

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Abstract—Confidence-weighted self-consistency beats majority voting when a frozen model's confidence is directionally calibrated. Every published scheme is monotone increasing in confidence, so an anti-correlated channel poisons the vote, and calibration gates survive inversion only by discarding signal. tact derives the vote exponent from the channel's measured signed within-item discrimination: a pooled van Elteren Somers' D, shrunk toward zero and mapped through a Bayes-discriminant link collapsing at base rate $\frac{1}{2}$ to $\gamma = z\sqrt{2+z^2}$. Inside the dead zone the vote is bit-identical to plain self-consistency. Assuming class-conditional noise, a label-free variant preserves the estimate's sign below a pair-weighted plurality-error rate of one half; beyond it, experiments show sign reversal. On a synthetic oracle it recovers channels pinning published protocols to the majority floor (1.000 vs. 0.762) and cracks the heterogeneity floor with zero paired losses (0.923 vs. 0.785); against a dev-picked signed grid the advantage narrows to distortion, echo and label-free operation. Two real-trace campaigns bound it: the channel is null on saturated benchmarks, positive on competition mathematics ($\widehat{D} = +0.250$), yet the stratum any such method can act on measures 2.5–7.5% of items across five substrates in two domains. Unless the pooled evidence clears its pre-specified threshold, the dead zone leaves the vote unchanged.

Index Terms—large language models, self-consistency, confidence calibration, label-free estimation, rank statistics

I. Introduction

Self-consistency (sc) [1] improves the reasoning accuracy of a frozen large language model (LLM) by sampling K chain-of-thought traces and returning the plurality answer. Each trace can also report a confidence score, verbalized [2], [3], derived from token log-probabilities, or elicited as $P(\text{True})$ [4], so weighting votes by confidence is a natural refinement. Confidence-Informed Self-Consistency (CISC) [5] showed that this recovers the accuracy of plain sc at a fraction of the sampling budget, and introduced Within-Question Discrimination (WQD) to identify discrimination as the confidence property relevant to voting, independently of calibration.

The methods surveyed here leave one structural case untreated. Their weighting rules are monotone increasing in confidence, including CISC's softmax weights, reliability-aware pseudo-counts [6], and warmup-thresholded filtering [7]. They tune the magnitude of positive weighting without searching for an anti-correlated channel. Such direction errors can arise when reinforcement fine-tuning distorts verbalized confidence or distribution shift reverses an in-domain signal. In the experiments reported here, a simple anti-correlated channel ($\kappa = -0.6$; Section III) drives confidence-weighted baselines from near-perfect accuracy to far below the majority-vote floor, while signed weighting recovers the signal. A binary dev-set gate that disables the channel when calibration error is high survives the inversion but discards discriminative information: a systematically under-confident yet perfectly ranked channel fails an ECE gate for reasons unrelated to voting utility [5], [8].

This paper frames the problem as estimating one scalar: the signed within-item discrimination of the confidence channel, and mapping that scalar, with its uncertainty, to a vote exponent.

Signed confidence tempering. tact votes with $w_i = \exp(\gamma\varphi_i)$, where φ_i is the standardized van der Waerden score of trace i's within-item confidence midrank. The exponent γ is derived from a pooled van Elteren Somers' D (equal to $2 \cdot \mathrm{WQD} - 1$) with an exact tie-corrected null variance and an item-clustered jackknife standard error, shrunk by positive-part James–Stein with a significance floor, then mapped through a Bayes-discriminant link. Two anchors are exact: inside the dead zone the vote is bit-identical to sc via a shared code path, and the log-value feature map reproduces CISC-power. Since φ uses only within-item ranks, the method is invariant to every strictly monotone distortion of the confidence scale, and under compression it beats the best exponent in the tested grid $\{0.25, 0.5, 1, 2, 4\}$ (1.000 vs. 0.963). Because that optimum sits at the largest exponent tried, the comparison applies only to the reported grid.

Label-free estimation and its boundary. The crowdsourcing lineage estimates reliability from cross-annotator covariance [9], [10]; one exchangeable channel from one model offers no such structure. tact estimates the signed discrimination from deduplication-weighted agreement pseudo-labels under a proven class-conditional-noise identity, $\mathbb{E}[\widehat{D}_g] = (1 - 2\bar{\rho})D$. For $\bar{\rho} < 1/2$, the expected estimate preserves the sign and attenuates its magnitude. A split-half inversion de-attenuates conservatively and sign-aware alarms return the method to sc at the identifiability boundary. Section IX measures the reversal that occurs after $\bar{\rho}$ crosses $1/2$.

Identifiability under heterogeneity. With i.i.d. per-item coupling and no observable covariate, per-item label-free adaptation is closed: any monotone use of an item's own agreement statistic collapses to plurality reinforcement, the observable sign opposes the truth on 96% of the plurality-wrong items with $|D_q| > 0.3$, the ones where a flip could win, and $\{\kappa > 0, \text{minority right}\}$ and $\{\kappa < 0, \text{plurality right}\}$ induce the same observable law. Indexed instead by an observable covariate, the same estimator run per group recovers each group's signed coupling with zero paired losses to sc (Section VI).

Falsification and measurement on real traces. Four rejection criteria were fixed before implementation. The strongest comparisons were the published dev-calibrated CISC protocol, whose tuned temperature already interpolates sc↔CISC, and a dev-picked signed exponent grid. The signed grid is stronger on the homogeneous sweep: tact trails it in the mid-range, while its advantages occur under distortion, confident echo, and label-free operation. Two real-trace campaigns and a five-substrate window measurement then bound the scope of label-free aggregation. The channel is null on saturated benchmarks ($\widehat{D} = -0.219$, $z = -1.24$) and positive on competition mathematics ($+0.250$, $z = +2.54$), so the premise holds where the model is uncertain; but the stratum such a method can act on is 2.5–7.5% of items across all five substrates, in two domains, and does not widen as items harden. On both real substrates the in-pool oracle remains below the pre-registered endpoint, and the dead zone therefore leaves the observed sc decisions intact.

II. Related Work

Confidence-weighted self-consistency. sc [1] treats sampled traces as i.i.d. votes. CISC [5] weights votes by softmax-normalized confidence with a temperature tuned on a labeled split, and its WQD metric makes the discrimination-vs-calibration point that also motivates this work; the rank-calibration line [8] reaches the same conclusion independently. Weighted variants [11] and early-stopping families [12], [13] refine the budget. Self-certainty [14] is the closest relative in spirit, being the one published selector that scores candidates by a rank-like quantity, but it ranks across candidates with a fixed positive orientation and is not evaluated here; reliability-aware pseudo-counts [6] and warmup-thresholded filtering [7] adapt online but only re-scale positive trust. None of these searches a negative exponent. The weight family can represent negative trust, but the published hyperparameter grids omit its sign. SignGrid-dev demonstrates this by opening the same c^γ family to negative γ and reaching the signed oracle across the negative half-axis. What no published protocol does is estimate that sign, with or without labels. CISC's tuned temperature already supplies dev-calibrated sc↔CISC interpolation. tact-dev adds signed estimation, rank invariance, and an analytic grid-free map.

Reliability estimation without labels. Estimating worker reliability from agreement is classical [9], [15], [16]; spectral meta-learners [10] and recent LLM ensemble work [17], [18] exploit covariance across multiple predictors. The setting here has one exchangeable channel from one model and only per-item vote structure. Correlated errors can invalidate agreement proxies in this setting. The proposed estimator therefore combines a quantified attenuation identity with conservative de-attenuation and explicit alarms.

Shrinkage and rank statistics. The estimator assembles classical parts: stratified rank statistics [19], the James–Stein positive-part estimator [20], effective-sample-size corrections [21], [22], and normal-scores discriminant analysis. The contribution is their combination with exact sc/CISC anchors and a signed vote exponent.

Honest sibling result. A preceding system in the same line of work (RLEV-VoI, redundancy-discounted voting with value-of-information stopping) was evaluated under the same falsification discipline and failed it, dominated everywhere by a simple deduplication baseline, and is reported as a negative result. Its post-mortem isolated the confidence dilemma studied here.

III. Problem Setup

A. Notation

Every symbol used in the paper is collected in Table VII (Appendix A), grouped by the section that introduces it. Items $q = 1, \ldots, Q$; item q has m_q sampled traces. Trace (q, i) yields an answer $a_{q,i}$ in a discrete set and a confidence $c_{q,i} \in (0, 1)$; correctness is $y_{q,i} = \mathbf{1}[a_{q,i} = a^*_q]$, unobserved at test time. Plain sc returns $\arg\max_A n_q(A)$ where $n_q(A)$ counts votes for answer A. CISC-power weights votes by $c^\gamma_{q,i}$ with a fixed $\gamma > 0$.

B. The confidence dilemma

The synthetic oracle draws, per item, traces from a cluster mixture with a latent correct answer and generates confidence as

$$c_{q,i} = \mathrm{clip}\big(\tfrac{1}{2} + \kappa\,(y_{q,i} - \tfrac{1}{2}) + \varepsilon_{q,i},\, 0.01,\, 0.99\big), \qquad (1)$$

with noise $\varepsilon \sim \mathcal{N}(0, 0.1^2)$ and coupling $\kappa \in [-0.6, 0.6]$. Fig. 1 maps the baseline landscape before the proposed method existed: unconditional weighting (CISC, $\gamma = 1$) collapses on $\kappa < 0$; an ECE gate never opens off the well-calibrated diagonal; a sign-corrected AUC gate over dev labels nearly saturates the homogeneous sweep. This pre-measurement limits prospective gains to monotone confidence distortion, covariate heterogeneity, small dev sets, and label-free operation. The evaluation focuses on those cells.

IV. TACT

A. Vote family

Within item q, let $R_{q,i}$ be the midrank of $c_{q,i}$ (ties averaged) and

$$\varphi_{q,i} = \frac{v_{q,i} - \bar{v}_q}{\sigma_q}, \qquad v_{q,i} = \Phi^{-1}\Big(\frac{R_{q,i}}{m_q + 1}\Big), \qquad (2)$$

where σ_q is the realized standard deviation of v within the item (the no-tie value is 0.62 at m=4 but 0.95 at m=40; a

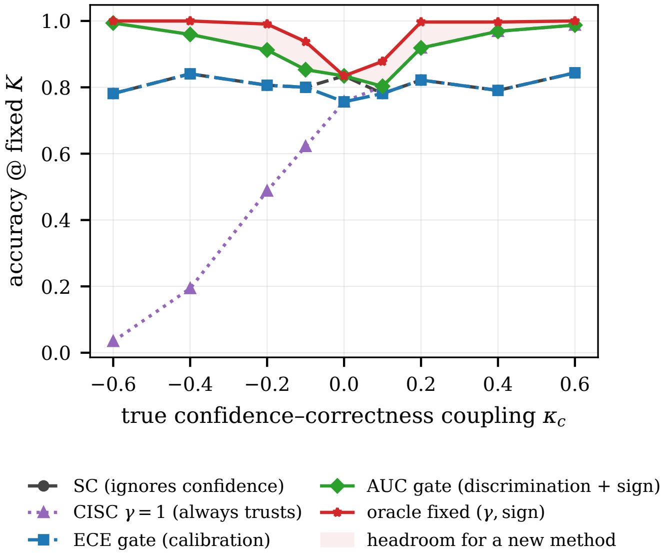

Fig. 1. The pre-measured problem statement: accuracy of baseline confidence policies at fixed K=15 as the true coupling κ varies. A trivial sign-corrected AUC gate (green) nearly saturates the homogeneous sweep; the headroom for any new method (shaded) concentrates in the mid-range and, off this plot, in distortion, heterogeneity, and label-free cells.

closed form would silently rescale γ across budgets), and $\varphi \equiv 0$ if $\sigma_q \le 10^{-8}$ (all-tied confidences vote as plain sc). The vote is

$$\hat{a}_q = \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\left(\gamma\, \varphi_{q,i}\right), \tag{3}$$

and when $\gamma = 0$ the implementation calls the sc routine itself, making the zero-trust anchor bitwise exact rather than equal in distribution. Because (2) depends on c only through within-item ranks, every strictly monotone distortion of the confidence scale leaves (3) unchanged.

B. Reliability statistic

For item q with n_q^1 positive and n_q^0 negative labels (dev: y; label-free: the pseudo-label of Section V), the Mann–Whitney statistic on midranks gives

$$D_q = 2\,\mathrm{AUC}_q - 1, \qquad \mathrm{AUC}_q = \frac{U_q}{n_q^1 n_q^0}, \tag{4}$$

which equals $2 \cdot \mathrm{WQD}_q - 1$ in CISC's notation. Pooling uses van Elteren pair-count weights $N_q = n_q^1 n_q^0$ [19]:

$$\widehat{D} = \frac{\sum_q N_q D_q}{\sum_q N_q}. \tag{5}$$

Under the within-item exchangeability null, U_q has the exact tie-corrected variance $n_q^1 n_q^0 (m_q{+}1)/12 \cdot [1 - \sum_t (t^3 - t)/(m_q^3 - m_q)]$, yielding a null standard error SE_0; between-item heterogeneity is captured by the closed-form delete-one-item jackknife SE_J. The conservative choice is

$$\mathrm{SE} = \max\left(\mathrm{SE}_0,\ \mathrm{SE}_J,\ \tfrac{1}{2\sqrt{N}}\right), \qquad r = \widehat{D}/\mathrm{SE}. \tag{6}$$

Because D is a pairwise functional, $\mathbb{E}[\widehat{D}]$ does not depend on m_q: an exponent estimated at m=40 transfers to deployment at m=8.

C. Tempering map

Shrinkage. Positive-part James–Stein with a significance floor ν:

$$\tilde{D} = \mathrm{sign}(\widehat{D})\, \max\bigl(0,\ |\widehat{D}| - \nu^2 \mathrm{SE}^2/|\widehat{D}|\bigr), \tag{7}$$

with dead zone $\{|r| \le \nu\}$; $\nu_{\mathrm{dev}} = 1.28$, $\nu_{\mathrm{LF}} = 2.33$. With $\nu = 1$, (7) is exactly the empirical-Bayes posterior mean under a $\mathcal{N}(0, \tau^2)$ prior with plug-in $\hat{\tau}^2 = \max(0, \widehat{D}^2 - \mathrm{SE}^2)$ [20]. The map is odd, continuous, never exceeds $|\widehat{D}|$, and is monotone in $\widehat{D}$ and anti-monotone in SE.

Link. Model $\varphi \,|\, y \sim \mathcal{N}(\mu_y, s^2)$ within item with the mixture standardized to unit variance, which is what (2) enforces, so $s^2 = 1/(1 + \bar{p}(1-\bar{p})u^2)$ where $u = \sqrt{2}\, \Phi^{-1}\bigl(\frac{1+\tilde{D}}{2}\bigr)$ and $\bar{p}$ is the base rate of correct traces. The Bayes-optimal per-trace log-weight coefficient is then

$$\gamma^* = \frac{u}{s} = u\sqrt{1 + \bar{p}(1-\bar{p})\, u^2}, \tag{8}$$

capped at $\gamma_{\max}$ (4 dev, 2 label-free). The uncorrected link $\gamma = u$ under-weights strong channels by up to $\sim$50% at $D = 0.9$. The link assumes φ is within-item normal, which (2) supplies only asymptotically: at m_q=4 the score takes four values before standardization. The scale consequence is handled by using the realized σ_q, but the distributional one is not, and it bites hardest in the small-budget setting this paper advertises (m=40 estimates transferring to m=8). Where $\widehat{D}$ saturates the cap binds and the link's shape is irrelevant; small m is where it has to hold and where it is least justified.

D. tact in one expression

Two simplifications collapse the pipeline. Factoring $\widehat{D}$ out of (7) makes the shrinkage a multiplicative gain in the pooled z-statistic $\zeta = \widehat{D}/\mathrm{SE}$ alone, and substituting $u = \sqrt{2}\, z$ with $z = \Phi^{-1}\bigl(\frac{1+\tilde{D}}{2}\bigr)$ into (8) removes the nested radical. tact is then

$$\boxed{\begin{aligned} \hat{a}_q &= \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\left(\gamma\, \varphi_{q,i}\right), \\ \gamma &= \Bigl[z\sqrt{2 + 4\bar{p}(1-\bar{p})z^2}\Bigr]_{-\gamma_{\max}}^{\gamma_{\max}}, \end{aligned}} \tag{9}$$

$$z = \Phi^{-1}\Bigl(\tfrac{1}{2}\bigl[1 + \widehat{D}\,(1 - \nu^2/\zeta^2)_+\bigr]\Bigr), \quad \zeta = \tfrac{\widehat{D}}{\mathrm{SE}}, \tag{10}$$

with $\widehat{D}$ from (5) and φ from (2). At the default $\bar{p} = \frac{1}{2}$ the exponent is exactly

$$\gamma = z\sqrt{2 + z^2}, \qquad z = \Phi^{-1}\bigl(\tfrac{1+\tilde{D}}{2}\bigr), \tag{11}$$

one probit and one square root, where $\tilde{D}$ is the shrunk pooled statistic of (7); it is distinct from the per-item AUC_q of (5). The map is not fitted to outcomes: ν is a significance level and $\gamma_{\max}$ a clip, both fixed before any data is seen. The clip is not cosmetic, though. Where $\widehat{D}$

Algorithm 1 tact: derive the exponent, then vote

Require: labeled dev pools $\mathcal{D}$, test pools $\mathcal{T}$, budget K, floor ν, cap $\gamma_{\max}$
Ensure: one scalar γ; an answer $\hat{a}_q$ for each $q \in \mathcal{T}$

$\mathcal{S} \leftarrow \emptyset,\ \ H \leftarrow \emptyset$
for $q \in \mathcal{D}$ **do**
  $y_i \leftarrow \mathbf{1}[a_{q,i} = a_q^*],\ \ i \le K; \quad H \leftarrow H \cup \{y\}$
  if $0 < \sum_i y_i < K$ **then** ▷ informative items only
    $R \leftarrow$ within-item midranks of $c_{q,1:K}$
    $D_q \leftarrow 2U_q/(n_q^1 n_q^0) - 1, \quad U_q$ from R
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{(D_q,\ N_q,\ \mathrm{Var}_0(D_q))\}$
$\widehat{D} \leftarrow \sum_q N_q D_q \,/\, \sum_q N_q$ ▷ van Elteren
$\mathrm{SE} \leftarrow \max\{\mathrm{SE}_0,\ \mathrm{SE}_J,\ 1/(2\sqrt{N})\}$
$\bar{p} \leftarrow \mathrm{clip}(\mathrm{mean}(H),\ 0.05,\ 0.95)$
$\zeta \leftarrow \widehat{D}/\mathrm{SE}$
if $|\zeta| \le \nu$ **then** ▷ dead zone
  $\gamma \leftarrow 0$
else
  $\tilde{D} \leftarrow \widehat{D}\,(1 - \nu^2/\zeta^2)$ ▷ positive-part JS
  $z \leftarrow \Phi^{-1}\big((1 + \tilde{D})/2\big)$
  $\gamma \leftarrow \mathrm{clip}\big(z\sqrt{2 + 4\bar{p}(1-\bar{p})z^2},\ \pm\gamma_{\max}\big)$
for $q \in \mathcal{T}$ **do**
  if $\gamma = 0$ **then**
    $\hat{a}_q \leftarrow \mathrm{SC}(a_{q,1:K})$ ▷ same routine, bit-identical
  else
    $\varphi \leftarrow$ standardized van der Waerden scores of $c_{q,1:K}$
    $\hat{a}_q \leftarrow \arg\max_A \sum_{i:a_{q,i}=A} e^{\gamma\varphi_i}$
return $\gamma,\ \{\hat{a}_q\}$

saturates it binds, and what reaches the vote is $\gamma_{\max}$, not the derived magnitude (Section VIII). The dead zone is now visible as a single condition, $|\zeta| \le \nu$, on which γ is identically zero and (9) is bitwise sc by Proposition 1. Equations (9)–(11) are verified equivalent to the shipped implementation over randomised inputs including every boundary (tests/test_formula.py).

Algorithm 1 states the labeled path end to end. Both loops cost $O(K \log K)$ per item, dominated by the within-item ranking, and the estimate is a single scalar: nothing item-specific crosses from dev to test, which is what makes the dead zone a global abstention.

E. Anchor properties

Proposition 1 (Exact sc reduction). At $\gamma = 0$, (3) equals plain sc as a function on every trace pool, including tie-breaks. Under $D = 0$, $P(\gamma = 0) \to 2\Phi(\nu) - 1$ (80% dev, 98% label-free), and γ is continuous through the dead-zone boundary, so a false positive applies an infinitesimal exponent.

Proposition 2 (Exact CISC reduction). With the feature map $\varphi_{q,i}^{\log} = \log c_{q,i} - \overline{\log c_q}$, the weights equal $\lambda_q\, c_{q,i}^{\gamma}$ with a per-item constant $\lambda_q > 0$ (distinct from the coupling κ

Algorithm 2 tact-LF: recovering the sign without labels

Require: pools $\mathcal{P}$, budget K, dedup threshold θ=0.95, margin quantile β=0.40, floor ν_{LF}, cap $\gamma_{\max}$, splits J=20, attenuation floor 0.20, minGated; $\hat{\eta}$ below is the estimated attenuation $1 - 2\bar{\rho}$, not $\bar{\rho}$
Ensure: γ, equal to 0 whenever any alarm fires

1: **for** $q \in \mathcal{P}$ **do**
2:   $w_i \leftarrow 1/|\mathrm{group}(i)|$ from single linkage at dup $\ge \theta$
3:   $M_q \leftarrow \arg\max_A \sum_{i:a_{q,i}=A} w_i$ ▷ dedup-weighted plurality
4:   $\mathrm{mgn}_q \leftarrow$ top-two dedup-weighted share gap
5: $G \leftarrow \{q : \mathrm{mgn}_q \ge Q_\beta(\mathrm{mgn}),\ \ge 2$ distinct answers$\}$
6: $A_{\mathrm{dup}} \leftarrow [\ \mathrm{median}_q(\text{Kish ratio}) < 0.5\ ]$ ▷ duplicate collapse
7: $A_{\mathrm{size}} \leftarrow [\ |G| < \text{minGated}\ ]$
8: $g_{q,i} \leftarrow \mathbf{1}[a_{q,i} = M_q]$ for $q \in G$
9: $(\widehat{D}_g, \mathrm{SE}_g, z_g) \leftarrow$ pooled statistic over G using g
10: $s \leftarrow \mathrm{sign}(\widehat{D}_g)$ ▷ estimated trust direction
11: $A_{\mathrm{mgn}} \leftarrow [\ \psi(s) > 0.05\ ]$ ▷ sign-aware margin decoupling
12: $\alpha \leftarrow$ mean two-half plurality agreement over J splits
13: $k \leftarrow$ inverse-Simpson size of the non-plurality mass
14: $p \leftarrow \big[1 + \sqrt{1 - (k{+}1)(1 - k\alpha)}\,\big]/(k{+}1)$
15: $A_{\mathrm{root}} \leftarrow [\ \text{discriminant} < 0.02\ ]$ ▷ root ambiguity
16: $\hat{\eta} \leftarrow \mathrm{clip}\big(\mathrm{UCB}_{95}(2p - 1),\ 0.20,\ 1\big)$
17: **if** $A_{\mathrm{dup}} \vee A_{\mathrm{mgn}} \vee A_{\mathrm{root}} \vee A_{\mathrm{size}}$ or $|z_g| \le \nu_{\mathrm{LF}}$ **then**
18:   **return** 0 ▷ refuse; the vote stays sc
19: **return** $\mathrm{Temper}\big(\widehat{D}_g/\hat{\eta},\ \mathrm{SE}_g/\hat{\eta}\big)$ with $\bar{p}$ unset

of (1)); hence the argmax, the ties, and the normalized vote shares coincide with CISC-power(γ) on every pool.

Proposition 3 (Regularity). The composite $g(\widehat{D}, \mathrm{SE})$ is continuous, odd, nondecreasing in $\widehat{D}$, nonincreasing in SE in magnitude, with $g(D, 0^+) = \gamma^*(D)$.

Proofs are elementary and pinned by unit tests in the released code (102 tests; the permutation-verified null variance, the EB identity in (7), and the link derivation (8) are each tested numerically).

V. Label-Free Estimation

A. Pipeline

(i) Dedup: single-linkage duplicate groups on the lexical-similarity channel at 0.95; each trace gets weight 1/|group| for plurality determination and pair weighting. (ii) Pseudo-label: $g_{q,i} = \mathbf{1}[a_{q,i} = M_q]$ with M_q the dedup-weighted plurality. (iii) Margin gate: keep the top 60% of items by dedup-weighted margin. (iv) Compute (5) with lab $= g$, giving $(\widehat{D}_g, \mathrm{SE}_g, r_g)$.

Two orderings in Algorithm 2 are load-bearing. The significance gate on line 17 tests the raw z_g, whose sign is unbiased by Proposition 4, while the tempering on line 19 uses the de-attenuated pair; testing the inflated statistic instead would let the de-attenuation manufacture significance. And $\bar{p}$ is left unset on the label-free path because the required base rate is unavailable without labels; the mixture correction of (8) is consequently omitted.

B. Sign consistency and its boundary

Proposition 4 (Attenuation identity). Let $\bar{\rho}$ be the pair-weighted probability that an item's plurality is wrong. If the plurality-error event is independent of φ given y (class-conditional noise), then $\mathbb{E}[\widehat{D}_g] = (1 - 2\bar{\rho})\, D$. In particular $\operatorname{sign} \mathbb{E}[\widehat{D}_g] = \operatorname{sign} D$ whenever $\bar{\rho} < 1/2$: the label-free estimate can only under-trust, never mis-sign.

The identity fails when the flip is caused by confidence, that is, under a confident echo. There the observable law under {majority right, $D < 0$} and {majority wrong via confident echo, $D > 0$} is identical (the two-root ambiguity of [10] restated for a single channel), so any label-free guarantee is necessarily conditional, and this paper states the condition.

C. De-attenuation and alarms

Split-half agreement over R=20 random half-splits estimates $\alpha = p^2 + (1-p)^2/k$ under a one-coin model with k effective wrong alternatives (inverse-Simpson), inverted as $p = [1 + \sqrt{1 - (k+1)(1 - k\alpha)}]/(k+1)$; $\widehat{D}_g$ is divided by the upper 95% bootstrap bound of $2p - 1$ (floored at 0.2), which can only under-inflate. Four semantically named alarms force $\gamma = 0$: A_{dup} detects duplicate collapse, A_{mgn} checks sign-aware margin decoupling, A_{root} marks ambiguity in the split-half quadratic, and A_{size} marks an insufficient gated set. The margin-decoupling alarm conditions on the estimated trust direction. An earlier sign-naive version ("plurality has the highest mean φ") false-alarmed on every benign anti-correlated channel; the released tests cover the corrected implementation. The significance gate acts on the raw z (unbiased sign under Proposition 4), whereas tempering uses the de-attenuated value. A semi-label-free mode takes only the sign from ∼50 dev labels, routing it into the pipeline and disabling the proxy-sign alarm. This removes the two-root ambiguity with a small labeled set.

VI. Heterogeneity: Impossibility and Escape

A. Per-item adaptation is closed under i.i.d. coupling

Suppose $\kappa_q \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.6^2)$ with no observable covariate.

Proposition 5 (Self-reinforcement). Any per-item rule $\gamma_q = h(\widehat{D}^g_q)$ with h monotone increasing and odd reinforces the plurality on both branches: $\widehat{D}^g_q > 0$ up-weights confident traces, which agree with the plurality; $\widehat{D}^g_q < 0$ up-weights unconfident traces, which are again the plurality side.

Remark 1. Measured in this harness, such a rule agrees with sc on 97.5% of items and its residual flips are net-harmful (1 right vs. 9 wrong per 400 items).

Remark 2 (Winner's curse). This is a measurement in the present harness, not a theorem: on plurality-wrong items with $|D_q| > 0.3$, the items where a flip could win, the agreement statistic's sign matches the true sign only 4% of the time.

Proposition 6 (Two-world unidentifiability). Let $w_1 = \{\kappa > 0, \text{ minority correct}\}$ and $w_2 = \{\kappa < 0, \text{ plurality correct}\}$. Computed against either truth, $D^{w_1} = -D^{w_2}$, so the statistic tact uses cannot order the two worlds. When the item has exactly two answer clusters the laws of (a, c) coincide outright and no label-free method can separate them. With three or more populated clusters they do not coincide: under w_1 only the correct minority carries elevated confidence, whereas under w_2 every non-plurality cluster does, so the conditional law of c on a third cluster separates them (verified numerically; the two laws agree on the top two clusters and differ with KS $p < 10^{-15}$ on the third). The impossibility is therefore conditional on the pool being effectively binary, which is the regime the confident-echo cell occupies: 88% of its items have no third cluster at K=15. tact does not exploit the residual signal, and no published method does either; doing so is left open.

Consequently the per-item oracle (0.973 in this harness) is unreachable, and the honest behaviour is to fall back to the global estimate, which tact's dead zone does: in the i.i.d. cell every variant returns bitwise sc (zero discordant pairs).

B. TACT-group

Real heterogeneity is typically indexed by an observable covariate (domain, question type). With κ indexed by a group label, running the estimator per group keeps every group inside the operating regime of Sections IV–V; groups with fewer than 30 dev (or 60 unlabeled) items fall back to the global estimate, which Propositions 5–6 show is the only defensible default.

VII. Experimental Setup

Harness. A cluster-mixture oracle generates, per item, up to $K_{\max}$=20 cached traces with answers, confidences (1), and two similarity channels; all methods replay identical pools (paired comparisons, exact McNemar tests). Voting budget K=15; 400 items per cell on the sweep, 600 for the group study; dev splits of 200 (primary) and 50 (small-dev).

Regimes. The κ sweep $\{-0.6, \ldots, +0.6\}$; three strictly monotone confidence distortions (compression toward 0.5, over-confident sigmoid, fourth power), rank-preserving by construction, so discrimination is intact while calibration is destroyed; i.i.d. heterogeneity ($\kappa_q \sim \mathcal{N}(0, 0.6^2)$); covariate-structured heterogeneity (three groups at $+0.6/0/-0.6$); and a confident-echo poison (a wrong cluster echoes verbatim with confidence 0.95).

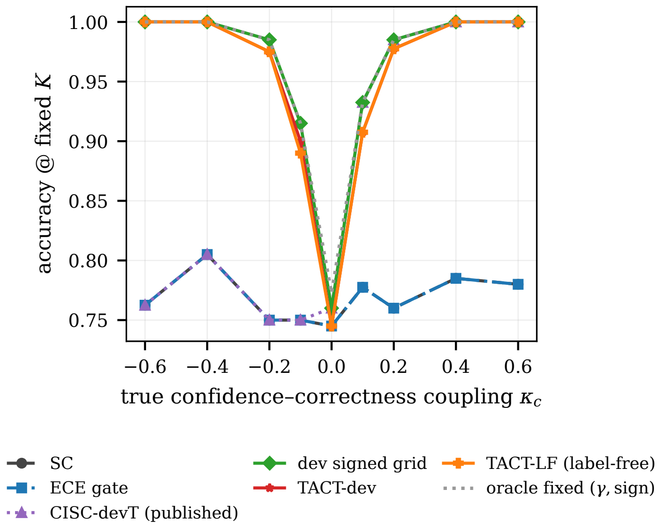

Fig. 2. Main result on the confidence-usage frontier. tact-dev and the fully label-free tact-LF track the signed oracle across the sweep; CISC-devT and the ECE gate sit at the sc floor for all $\kappa < 0$.

Baselines. sc; CISC-power with $\gamma \in \{0.25, \ldots, 4\}$; CISC-devT, the published dev-calibrated protocol (positive grid picked on dev); a binary ECE gate; SignGrid-dev, the strongest trivial baseline (signed exponent grid picked on dev); and the test-set oracle over signed fixed exponents as the upper envelope. The group study adds the naive self-referential per-item method as a negative control and the per-item link oracle as the ceiling.

Pre-registered rejection criteria. The decision rule is stated here because the protocol is offered as a contribution. Each criterion uses an exact paired McNemar test on the 400 items of a cell, at $\alpha = 0.05$ one-sided, with the seed-level bootstrap of Section VIII-F as the second gate: a falsifier fires when the single-cell test has $p < 0.05$ and the across-seed interval excludes zero. Single-cell tests at this size are underpowered against a strong baseline, which is the reason for the second gate; the earlier fixed $\tau = 0.02$ tolerance is retained only as a reporting convenience in the tables. Rejection would occur if tact-dev fell more than τ below the best fixed-γ CISC at $\kappa = +0.6$, if either variant fell more than τ below sc anywhere on the sweep, or if tact-LF failed to beat the ECE gate on sweep average. The strongest criterion asks whether CISC-devT or SignGrid-dev stays within τ of tact-dev in every distortion, heterogeneity, and small-dev cell. No single-cell comparison with SignGrid-dev rejects equality on the homogeneous sweep (the smallest p is 0.08 at $\kappa = +0.1$). The ten-seed bootstrap nevertheless resolves a systematic mid-range deficit, which a single-seed analysis would have reported as a tie.

TABLE I

Coupling sweep (accuracy at K=15; 400 paired items per cell; dev n=200). Published protocols sit at the sc floor on the entire negative half-axis.

κ	sc	ECE	devT	SignGrid	tact-dev	tact-LF	oracle
−0.6	.762	.762	.762	1.000	1.000	1.000	1.000
−0.4	.805	.805	.805	1.000	1.000	1.000	1.000
−0.2	.750	.750	.750	.985	.975	.975	.985
−0.1	.750	.750	.750	.915	.900	.890	.915
0.0	.745	.745	.760	.760	.745	.745	.772
+0.1	.777	.777	.932	.932	.907	.907	.932
+0.2	.760	.760	.985	.985	.978	.978	.985
+0.4	.785	.785	1.000	1.000	1.000	1.000	1.000
+0.6	.780	.780	1.000	1.000	1.000	1.000	1.000

TABLE II

Adversarial regimes (accuracy at K=15). "Oracle" is the test-set best over raw-value weight policies; rank invariance beats that entire family under compression.

Regime	sc	devT	SignGrid	tact-dev	tact-LF
Monotone compress	.775	.963	.963	1.000	1.000
Monotone overconf	.775	1.000	1.000	1.000	1.000
Monotone power	.775	1.000	1.000	1.000	1.000
Hetero (i.i.d.)	.765	.765	.765	.765	.765
Confident echo	.190	.190	.568	.615	.190†

†alarm fires and the method refuses to leave sc: the conditional guarantee of Prop. 4 working as stated.

VIII. Results

A. Signed recovery, with and without labels

Table I and Fig. 2 give the sweep. On $\kappa < 0$, the published protocols remain at the sc floor because CISC-devT searches only positive exponents and the ECE gate stays closed. Dev ECE ranges from 0.10 to 0.80 even though discrimination is perfect at the extremes. The label-free and 200-label variants are nearly coincident. At $\kappa = -0.6$, for example, the raw agreement statistic is $\widehat{D}_g = -0.81$ with $z = -17.6$, and tact-LF reaches 1.000 without labels. At $\kappa = 0$, the dead zone sets $\gamma = 0$ exactly, producing the same decisions as sc on every paired item.

Where the derived exponent actually operates. The four cells that carry the headline 1.000 ($\kappa = \pm 0.4, \pm 0.6$) are cells where $\widehat{D}$ saturates, so the link returns an untempered γ^* between −8.4 and +12.1 across the seven saturated cells and the cap $\gamma_{\max} = 4$ is what the vote actually sees. Performance in those cells depends on recovering the sign. At $\kappa = \pm 0.1, \pm 0.2$, where the derived value lands strictly inside the cap ($|\gamma|$ from 1.10 to 2.91), tact-dev trails the dev-picked signed grid on all four cells (0.900 vs. 0.915; 0.907 vs. 0.932; 0.975 vs. 0.985; 0.978 vs. 0.985). The large advantage over the published protocols is attributable to signed weighting. On these mid-range cells, the analytic map selects a less effective magnitude than the signed grid. Its advantage over the grid occurs under monotone distortion, confident echo, and label-free operation. The one place the interpolation itself pays is confident echo,

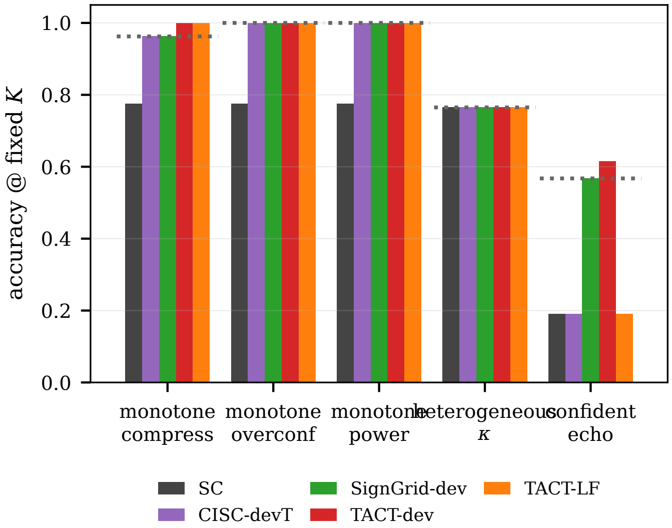

Fig. 3. Adversarial regimes. Dotted line: the oracle over raw-value weights. Left group of bars: rank invariance beats that family under compression; right: the labeled variant counters the confident echo while the label-free variant alarms and refuses.

where $\gamma = -1.198$ falls between grid points and beats the grid optimum $\gamma = -1$ (0.615 vs. 0.568).

B. Rank invariance where raw values fail

Under monotone compression (Table II, Fig. 3) all confidences clustered near 0.5, so every c^{γ}-family weight was nearly uniform: even the oracle over raw-value policies reached only 0.963. tact's rank scores were unchanged by the distortion and both variants reached 1.000. Under the confident echo, dev labels showed that high confidence predicted an incorrect trace. tact-dev responded with $\gamma = -1.20$ and reached 0.615, compared with the sc floor of 0.190. On the label-free path, the duplicate-collapse alarm fired and left the vote at sc. Proposition 6 explains this result: 88% of the items are effectively binary, so their observed pools do not identify the sign.

C. Heterogeneity

Table III and Fig. 4 give the group study. In the covariate-structured cell, per-group tact recovers each group's signed coupling: the dev exponents are $\{+4.0, 0.0, -4.0\}$ and the label-free exponents are $\{+2.0, 0.0, -2.0\}$, with the $\kappa = 0$ group inside the dead zone. The label-free variant reaches 0.923, within 0.023 of the per-item link oracle, and records no paired losses to sc over 600 items ($+83/-0$, $p = 2.1 \times 10^{-25}$). In the i.i.d. cell every method remains at the floor with zero discordant pairs, including the naive self-referential control. That control reaches 0.787 against the 0.785 floor in the grouped cell, changing only two items in 600 despite access to the same covariate used by the per-group estimator. The two tact arms land within seed noise of each other here (0.923 and 0.927 on one seed; 0.929 ± 0.015 each over five). A natural explanation would be the arms' different exponent

TABLE III
Heterogeneity study (600 paired items; K=15).

Method	Grouped	i.i.d.
sc (floor)	.785	.752
tact global (dev)	.785	.752
tact-group (dev)	.927	.752
tact-group (label-free)	.923	.752
Naive per-item (neg. control)	.787	.752
Per-item link oracle (ceiling)	.947	.973

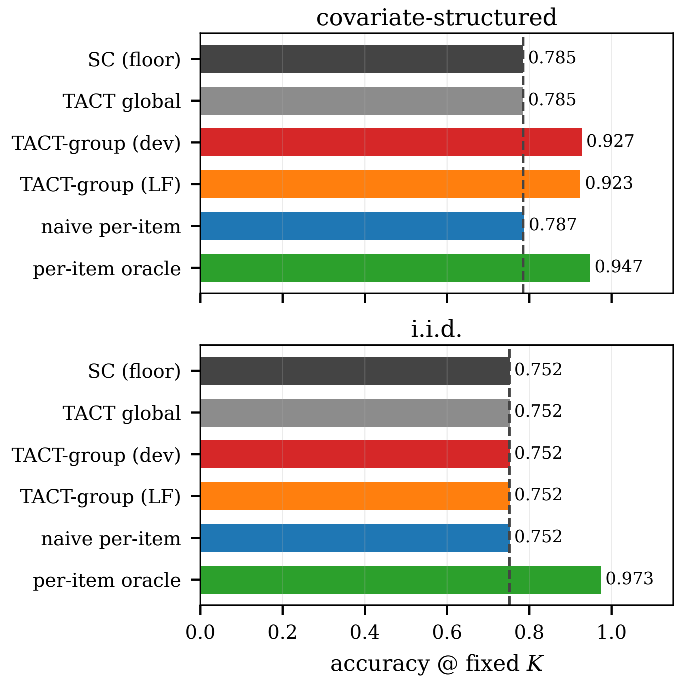

Fig. 4. Structured vs. i.i.d. heterogeneity. Left: with an observable covariate, per-group tact (label-free) approaches the per-item oracle from the 0.785 floor with zero losses to sc. Right: every method, including the negative control, remains at the floor in the closed i.i.d. cell.

caps; the ablation rules it out. Sweeping the cap over $\{1, 2, 3, 4, 6, 8\}$ moves neither arm at all (spread 0.0000 for both), so the cap is not load-bearing in this cell and the difference is sampling variation.

D. Small dev sets and falsifiers

With dev n=50 the conclusions are unchanged (1.000 at $|\kappa|$=0.6; 0.978 at -0.2): the SE-aware shrinkage degrades smoothly. The pre-registered rejection criteria remain unmet. At $\kappa = +0.6$, tact-dev and the best fixed-γ CISC both score 1.000. At $\kappa = 0$, tact is bit-identical to sc, and it stays within the 0.02 tolerance everywhere else. tact-LF has a sweep mean of 0.944 against 0.768 for the ECE gate. Both grid baselines trail in the distortion and echo cells by 0.037 and 0.047, respectively, and neither operates without labels. On the single seed the paired tests against SignGrid-dev do not reject equality anywhere (κ=-0.2: 3/7 discordant, exact $p = 0.34$; κ=$+0.1$: 8/18,

TABLE IV
What the test suite verifies. Every proposition in the paper has an executable counterpart; the counter-tests fail deliberately on rejected alternatives so a regression cannot silently reinstate them.

Claim	Evidence
Prop. 1 (exact sc)	identical incl. ties, 200 pools; dead-zone rate >70% under D=0
Prop. 2 (exact CISC)	identical vote shares, 100 pools
Prop. 4 (attenuation)	$\rho \in \{.1, .25, .4\}$, abs. .06
Props. 5–6	97.5% sc agreement; 4% sign match; the two-world boundary is pinned both ways (binary pools indistinguishable, a populated third cluster separates them at KS $p < 10^{-10}$)
Rank invariance	3 distortions × 100 pools
Estimator internals	permutation null variance (3,000 draws, 10% tol.), JS–EB identity to 10^{-12}, link (8) to rel. 10^{-9}, permutation-invariance regression
Rejected alternatives	Kish ESS and the SAFE-under-VoI guarantee each have a test asserting their failure

$p = 0.08$), and the ten-seed intervals of Section VIII-F are what establish the mid-range deficit. Against SignGrid-dev the margin is narrow on the homogeneous sweep. tact trails by 0.007–0.025 in the mid-range, consistent with the cost of shrinkage. Its net advantages are concentrated in distortion (+0.037), echo (+0.047), and label-free operation, which the grid cannot perform.

E. Verification of the implementation

The released code gives each mathematical claim in Sections IV–VI an executable counterpart; the suite contains 102 tests covering tact and the follow-on work. Table IV maps propositions to the tests that would fail if they stopped holding.

The permutation-invariance test was added after a memoisation key allowed the test to pass while the estimator itself was order-dependent by up to 0.10; it now calls the internal routine directly. The last two rows are counter-tests that assert failure of rejected alternatives: the Kish effective-sample-size formulation and the claim that the shipped default honours the SAFE stopping guarantee. A later change that reinstates either alternative will therefore fail the suite.

F. A harness artifact, and dispersion across seeds

Two corrections to the synthetic results, both found by re-running what had been single-seed measurements.

Tie-breaking was rewarding sc for free. The generator assigned the correct answer the code 0 on every item, and argmax breaks ties toward the lowest index, so on any item whose vote was tied plain sc chose correctly by construction: 67.7% on near-tied items against 50.4% for a random tie-break, with 31.8% of items tied at K=15. Every method that perturbs the weights off integers forfeits that subsidy, so the artifact inflated the baseline and penalised the proposed method. It also produced a spurious falsifier: at $\kappa = 0$, tact-dev averaged 0.797 against sc at 0.818 across ten seeds, violating the registered sc-safety criterion, while the exponents responsible were $|\gamma| \leq 0.05$. The items were near-ties whose tie-break had moved, not items the exponent had reweighted. Answer codes are now permuted per item. With the artifact removed the two coincide exactly where the dead zone should hold: $\kappa = 0$ gives 0.745 for both methods and heterogeneous-κ gives 0.765 for both (ten-seed means 0.758 and 0.768, zero discordant pairs in every seed): the dead zone behaves as Proposition 1 states. All synthetic numbers in this paper are post-fix.

Dispersion. Ten seeds per cell, 400 paired items each, bootstrap over seeds. The extremes are stable to the third decimal (1.000 ± 0.001 at $|\kappa| \geq 0.4$). The mid-range deficit against SignGrid-dev is small but real, not noise: -0.013 $[-0.017, -0.009]$ at $\kappa = -0.2$, -0.016 $[-0.024, -0.009]$ at -0.1, -0.012 $[-0.018, -0.005]$ at $+0.1$, -0.013 $[-0.019, -0.008]$ at $+0.2$, all $p < 0.001$. So is the advantage where the paper claims it: $+0.032$ $[+0.027, +0.037]$ on monotone compression and $+0.048$ $[+0.037, +0.059]$ on confident echo. The one cell that changes sign under the fix is heterogeneous-κ, now $+0.011$ $[0.000, +0.031]$, where it had been a loss. Reporting a single seed would have hidden both the artifact and the fact that the mid-range gap is systematic.

G. Real-trace validation

Validation on real traces used Claude Haiku 4.5 as the frozen model: 100 items, 50 from GSM8K [23] and 50 from CommonsenseQA, with 12 independent chain-of-thought traces and a verbalized confidence per item (1,200 traces total), evaluated at K=12 with a 40/60 dev/test split. The results revealed both a calibration failure and a limitation imposed by the substrate.

Calibration and discrimination diverge on real data. The channel is extremely well calibrated in the usual sense: ECE = 0.016, far inside the 0.10 gate, so a binary ECE gate opens and hands the channel to CISC. Yet the measured within-item discrimination is $\widehat{D} = -0.219$ with SE = 0.176 ($z = -1.24$): no usable signal, and what little there is points the wrong way (math -0.515, commonsense -0.173; both groups negative). This is the exact mirror image of the synthetic case in which ECE wrongly closed the gate on a discriminative channel (Section III): on real traces ECE wrongly opens it on a non-discriminative one. Calibration is uninformative about voting utility in both directions, and a signed discrimination statistic is what distinguishes them.

The dead zone leaves every decision unchanged. With $|z| < \nu$, tact-dev, tact-LF and tact-group all return $\gamma = 0$ and are bit-identical to sc on every test item ($+0/-0$ discordant pairs, $p = 1$). All methods score 0.917. This is the pre-registered null-direction prediction of Section IX confirmed on real data: where the channel carries no signal, the method is free.

Saturation limits the usable sample. Trace-level accuracy is 0.958 on GSM8K and 0.847 on CommonsenseQA, so only 12 of 100 items contain both a correct and an incorrect trace, the only items a within-item rank statistic can use. The estimator is not underpowered by design; the benchmark simply does not present the model with enough genuine uncertainty. Exposing non-null coupling on a strong model requires harder item pools, not more traces per item.

Verbalized confidence is tie-heavy. Two values (0.99, 0.95) account for 49% of all reports, activating the tie-safe degeneration path of (2) on many items.

The abstention is measurable but fragile in this campaign. A fixed-zero rule reproduces the dead-zone result exactly, so the campaign alone cannot distinguish a working safety mechanism from an absent signal. Replays of the cached pools provide two additional checks. In an item bootstrap, the dev gate opens in 50.4% of resamples. The observed $|z| = 1.24$ lies only 3% below $\nu = 1.28$, so the abstention is close to the threshold. The label-free significance gate opens in 16.2% of resamples, although the planted-sensitivity replay below showed that another structural gate prevents the full method from acting. Permuting confidences within items preserves the real marginals, ties, pool composition and difficulty and destroys only the coupling, puts the false-open rate at 18.4% against a nominal 20% (1.6% against 2% label-free), with the null z at mean -0.002 and SD 0.954. This real-substrate null calibration provides direct evidence about the dead zone's specificity. To measure the cost of the observed decision, overriding the gate, the derived $\gamma = -0.40$ would have scored 0.933 ($+1/-0$) and a gold-label oracle reaches 0.950 at $\gamma = -1.25$, a ceiling of two items. Neither approaches significance, so the substrate is null in the way that matters. The estimator and oracle nevertheless agree on the sign, so the zero-cost result is limited to what can be measured at this sample size. This result did not establish that abstaining was right.

A planted channel measures sensitivity. The bootstrap and permutation replays measure stability and specificity. Sensitivity requires a channel of known strength, so one is planted into these same pools. Real answers, difficulty, and duplication are unchanged; only the confidence vector is replaced, conditioned on the real correctness pattern. Planting only the estimator's own location-shift model would prove nothing (the invariance-group pitfall of Section IX), so four families are swept: the Gaussian location shift; the same shift under Laplace noise, violating the link's normality assumption while preserving D; a rank-concentrated channel that forces the top-confidence slot onto a correct trace; and a variance-only channel, informative but unsigned. Across 40 seeds per cell, the gate's operating characteristic follows the realized $|\widehat{D}|$ across the Gaussian and Laplace families. It reaches half-power near $|\widehat{D}| = 0.247$ (Gaussian) and 0.208 (Laplace), full power by 0.49 and 0.56, and the derived exponent capturing the whole oracle gain from 0.83 and 0.95. At the Gaussian thresholds the Laplace family is already at 0.975 power and 0.98 capture. Detection therefore follows the rank functional across this normality violation. The measured channel above, $|\widehat{D}| = 0.219$, lies near the half-power point, consistent with the bootstrap result. Whenever the gate opens, its sign is correct in every tested cell.

The experiment also exposes two scope boundaries. The rank-concentrated channel yields $|\widehat{D}| \leq 0.183$ at full strength. A pooled pair statistic barely detects information carried by a single rank position and leaves the derived gain near zero where a gold-label exponent takes $+0.039$; the unsigned channel is invisible by construction and the gate holds its nominal false-open rate on it. The label-free arm revealed a separate defect: it returned $\gamma = 0$ in every cell, even on a saturating channel with the item floor lowered to one, because with 85 of 100 items unanimous the margin gate's 40%-quantile cut equals 1.0, and no item satisfies margin ≥ 1.0 with two distinct answers. Its abstention in the real-trace campaign was therefore structural rather than a reading of the channel, and on saturated substrates the quantile gate needs an absolute cut.

Scope of this first campaign: one model, two benchmarks, K=12. It confirmed the null-direction prediction and the calibration–discrimination argument, and it is not evidence that tact improves accuracy, since the channel carried no signal to exploit. The saturation diagnosis motivates the harder campaign in Section VIII-H.

H. Confirmatory campaign on harder items

The saturation diagnosis predicts that a channel measured as null on saturated benchmarks should become measurable on items the model finds genuinely uncertain. A pre-registered follow-up tests that prediction: 119 MATH level-5 problems [24], [25], 16 traces each from the same frozen model, and a 30-item sign set and an 89-item evaluation set drawn from the registered list before any trace was collected. The registration specified the expected discrimination sign, a paired accuracy endpoint, baseline and safety checks, and substrate diagnostics.

The confidence channel is detectable. On the evaluation set the pooled statistic is $\widehat{D} = +0.250$ with SE = 0.098, so $z = +2.54$ clears the registered $z \geq 2$ threshold. This is the first real-trace evidence that verbalized confidence carries positive within-item discrimination; the same measurement on GSM8K/CommonsenseQA gave -0.219 ($z = -1.24$).

The realized substrate nevertheless saturated again: per-trace accuracy 0.819, sc 0.888, a decisive stratum of 10 of 89 items, and the correct answer present in the pool on only 4 of those. The in-pool oracle therefore tops out at $+4/-0$, with exact one-sided $p = 0.0625$. Even this oracle missed the registered paired-accuracy endpoint,

TABLE V
Confirmatory campaign, MATH level-5 evaluation set (89 items, K=16). Every method replays the same cached pools. The duplication channel is inert because no reasoning text was collected, so dedup-sc coincides with sc by construction.

Method	Accuracy	net vs. sc
sc / dedup-sc / CISC-linear	0.888	n/a
tact-LF	0.888	$+0/-0$
tact-semi-LF	0.888	$+0/-0$
best-single-confidence	0.843	$+1/-5$
in-pool oracle (ceiling)	0.933	$+4/-0$

establishing that the realized pool cannot support a detectable aggregation gain at the registered level.

Both label-free paths returned $\gamma = 0$, although for different reasons. The fully label-free arm was stopped by A_{size} and A_{mgn}: only 13 items reach the margin gate with a pseudo-label that varies, against a floor of 50, and relaxing the gate's quantile all the way to 0 raises that only to 39, so the arm is untestable here at any channel strength. The semi-label-free arm was stopped by its 30-item sign set reached $z = +0.83$ against a $|z| > 1$ gate, while the evaluation set on the same substrate reached $z = +2.54$. This mismatch indicated limited power in the sign set; increasing its size directly addresses the shortfall. Both paths are therefore bit-identical to sc at 0.888, satisfying the registered baseline and safety checks. The cost of acting anyway is visible in the same table: best-single-confidence, which uses the channel on every item, loses 4.5 points at 0.843. A_{size} reports item supply; it carries no estimate of the confidence channel. The planted-sensitivity replay showed two ways it can fire: on the saturated pools of Section VIII-G the margin gate's quantile cut degenerates to 1.0 and the gated set is empty for any channel, whereas here the cut is 0.875 and the set is non-empty, so the alarm reports a genuine shortfall against the floor. The quantile degeneration in saturated pools is the defect; the present MATH result is a genuine supply shortfall.

One caveat from this campaign transfers beyond tact. Measured difficulty depended on the collection protocol: a 30-problem-per-call probe put level-5 plurality accuracy at 0.40, while the 15-problem-per-call confirmatory run yielded 0.888 on the same stratum. Batch size belongs in the experimental record whenever traces are collected in batches.

I. How wide is the addressable stratum?

Both campaigns failed their endpoint for the same reason, which suggests measuring that reason directly. Define the window as the fraction of items where the plurality is wrong and the correct answer is present in the pool: the ceiling for any label-free aggregation method, since nothing outside it can be changed.

The window was measured on five substrates spanning two domains (Table VI). For code generation, where an executable test suite supplies per-sample ground truth and the window might reasonably be expected to widen, 40 LeetCode Medium/Hard problems [26] were solved 8 times each and graded against the benchmark's hidden suites, with the baseline taken as the largest behavioural cluster over probe inputs without comparing them with expected outputs. The window is $3/40 = 7.5\%$ (CI_{95} 2.6–19.9%): wider than label-free QA, but the same order, and the composition is the same shape at 30 saturated, 7 capability wall and 3 rescuable, i.e. 75/17.5/7.5%. Nor does budget open it. The seven capability-wall problems produced zero correct solutions in 224 further attempts (per-problem 95% upper bound on the pass rate 0.088), and extrapolating oracle@N showed the window saturating by N=32.

TABLE VI
The addressable stratum across substrates, on one definition throughout: decisive is the fraction of items whose plurality is wrong, window (Win.) the fraction that are decisive and have the correct answer somewhere in the pool; both in per cent. The window is the ceiling for any label-free aggregation method. Rows marked † are measured here; HumanEval+/MBPP+ is recomputed from published oracle-minus-selector tables, which report the window only. As items harden the decisive fraction grows but the window does not: they pass from saturated to capability-limited. The MATH L5 row is over the full registered list of 119 items, the substrate being the object measured; Section VIII-H quotes the same quantities over the 89-item evaluation set that remains after the sign set is split off (10 and 4 items, so 11.2 and 4.5). The last row is that same MATH L5 substrate under the budget manipulation of Section VIII-J, not a sixth substrate: the 2.5–7.5% range quoted throughout is over the five above it.

Domain	Substrate	Dec.	Win.
QA	GSM8K / CSQA†	9.0	4.0
QA	MATH L5†	7.6	2.5
QA	AIME / AMC†	23.3	3.3
Code	HumanEval+ / MBPP+	n/a	3.6
Code	LeetCode Med/Hard†	25.0	7.5
QA, budget-capped	MATH L5†	18.5	11.8

The grading harness was validated against the benchmark's own reference solutions before any candidate was scored: 178 of 180 pass under the sandbox's resource limits. The check is not a formality: a sandbox whose resource limits the host rejects outright fails 100% of executions, and that condition appeared as a candidate failure. Studies that grade by execution should report their reference-solution pass rate for the same reason a calibration curve is reported. Without this check, candidate failures cannot be attributed to model capability because the sandbox can produce the same observed outcome.

J. A third axis: reasoning budget

Difficulty is not the only way to lower per-sample accuracy. Constraining the reasoning budget lowers it while leaving the correct answer inside the model's competence, which is the combination the decisive stratum needs and that difficulty does not supply: as items harden they pass from saturated to capability-limited without pausing in between. A paired run tests this on the same 119 items, same model, same K, with the model instructed to answer without writing any working.

The manipulation was weaker than pre-registered on its stated check (per-sample accuracy $0.853 \rightarrow 0.781$, against a 0.10 drop required) and the correct answer did not leave the pool ($0.950 \rightarrow 0.933$), so what follows is exploratory. It is reported because the constraint did not lower accuracy uniformly, it moved items across the plurality boundary: sc fell $0.924 \rightarrow 0.815$, the window widened from 2.5% to 11.8% (paired exact McNemar on window membership, 13 in and 2 out, $p = 0.0037$), and the channel grew stronger, $\widehat{D}$ from $+0.229$ ($z = 2.69$) to $+0.398$ ($z = 6.76$). Forced to answer without working, the model's confidence tracks whether it happened to be right.

This is the widest window measured anywhere in this paper, and the first substrate on which the in-pool oracle clears the endpoint at all ($+9/-0$, $p = 0.002$ on a decisive stratum of 17). tact nonetheless returned $\gamma = 0$: only 12 items survived the margin gate against the threshold of 30, so A_{size} fired. This is the same item-supply condition, with the quantile cut at 0.875 and the gated set again non-empty. It differs from the degenerate quantile case in the planted-sensitivity replay. That abstention is testable, and it cost nothing. Estimating γ from gold labels, an upper bound no deployable method has access to, the best available is $+4/-1$ at $\gamma = 1$ ($p = 0.19$); the derived $\gamma = 0.670$ gives $+3/-1$. The gap that matters is therefore not the one the window closes. Even with the window at 11.8% and an oracle able to convert nine items, confidence weighting reaches fewer than half of them, and none of the reachable configurations reaches $p < 0.05$.

IX. Discussion and Limitations

What the evidence does and does not show. The accuracy claims are all on a synthetic oracle whose confidence model (1) is, at the homogeneous cells, the very coupling the estimator measures. Three design choices limit the circularity, and the first is weaker than it looks: heterogeneity and echo lie outside the estimator's working model, but the three distortion cells are rank-preserving by construction and therefore sit inside tact's own invariance group, so passing them shows only that the implementation respects an invariance it was built to have; mechanism-recovery claims (does $\widehat{D}$ track κ?) are reported separately from accuracy claims; and the pre-measured baseline landscape (Fig. 1) fixed the winnable cells before the method existed. The real-trace campaigns of Sections VIII-G and VIII-H test the premise and the abstention behaviour, and both predictions held: the channel is null on saturated benchmarks and positive on competition mathematics, and the dead zone kept the vote bit-identical to sc in each case. They do not test the accuracy claim, because on neither substrate was the addressable stratum large enough for any method to demonstrate a gain (Section VIII-I).

Narrow margins where labels abound. When labels are plentiful and the confidence scale is trusted, a dev-picked signed grid captures most of the value; tact's case rests on the label-free setting, distorted scales, small dev sets, and the exactness of its anchors.

Conditional label-free guarantee, and what happens past the boundary. Proposition 4 requires $\bar{\rho} < 1/2$ after deduplication, and the confident-echo ambiguity is fundamental (Proposition 6). Follow-on work measured the consequence of crossing that boundary, and it is worse than under-trust. In a paraphrased wrong-majority cell (a dominant wrong cluster that is semantically tight but carries no verbatim signature, so deduplication has nothing to collapse) the plurality is wrong on most items, $\bar{\rho} > 1/2$, and tact-LF does not merely shrink toward sc: it mis-signs, saturates at $\gamma = -2.0$, and scores 0.000 against an sc floor of 0.340. None of the four alarms fires, because A_{dup} keys on verbatim duplication which is absent by construction. This is the method's sharpest unguarded failure mode: the guarantee is conditional, the condition is not observable label-free, and the existing diagnostics do not detect its violation. Where a systematically wrong majority is plausible, the semi-label-free mode uses $\sim$50 labels to identify the sign; the fully label-free path has no corresponding guard. A second possible remedy is semantic clustering. It is not evaluated here, although the failure is driven by a semantically tight wrong cluster that lexical deduplication cannot see, which is precisely what semantic-equivalence clustering [27] is built to collapse. Substituting a semantic pseudo-label for the lexical one is the obvious next guard, and this paper does not test it.

Global exponent per group. Within a group, tact ships one exponent; per-item variation inside a group is unexploitable by Propositions 5–6 unless further covariates exist.

The thin-window phenomenon. Section VIII-I measures the stratum this whole family of methods can act on at 2.5–7.5% of items across all five substrates, in two domains, with no widening as items harden: they pass from saturated straight to capability-limited. At these widths, the real campaigns provide no evidence for changing the sc decision. Best-single-confidence, which uses the confidence channel on every item, loses 4.5 points in Table V, while the dead zone holds tact at the sc floor. A synthetic gain of the reported size is also not measurable on a benchmark of a few hundred items. The real-trace claim is therefore limited to signed channel measurement and the observed abstention behaviour. Demonstrating an accuracy gain needs a (model, benchmark) pair whose plurality is wrong on 30–60% of items with the correct answer still reachable, and no pair tried here satisfies both. Section VIII-J widens the window to 11.8%, with an oracle able to convert nine items, yet confidence weighting reaches fewer than half of them. Both window width and conversion efficiency constrain the measured gain.

Widening the window on purpose. That result also names the most promising direction the paper does not pursue. Difficulty moves items from saturated straight to capability-limited, but the budget constraint of Section VIII-J moved them across the plurality boundary while leaving the answer inside the model's competence, which is the combination the decisive stratum needs; sampling temperature is a second knob with the same character, and neither was swept. If the sampling regime is a controllable input, not a fixed property of the substrate, the question changes from "how wide is the window?" to "what decoding configuration puts the most items in it, at what accuracy cost, and does the confidence channel stay signed there?" The budget arm answers the first part once and suggests the channel gets stronger under the constraint ($\widehat{D}$ from +0.229 to +0.398), which would be worth confirming on more than one substrate before it is believed. A method that chose its own decoding budget to maximise decisive-stratum mass, then aggregated, would be a different contribution from this one, and the measurement here is what makes it worth attempting.

X. Conclusion

tact estimates a signed, uncertainty-aware scalar for confidence-weighted self-consistency and can recover its sign without labels under the stated class-conditional-noise condition. Across five substrates in two domains, only 2.5–7.5% of items fall within the stratum where label-free aggregation could change an incorrect plurality to the correct in-pool answer. Neither real-trace campaign provides enough such items for the in-pool oracle to clear the pre-registered accuracy endpoint. In both campaigns, the dead zone sets $\gamma = 0$ and reproduces the corresponding sc decisions.

Code and Data Availability

All code, cached traces, and the JSON artifacts behind every table are at https://github.com/vito1317/adaptive-reasoning-consensus. Table IV is produced by pytest at commit 5f14b20; the synthetic results by experiments/run_tact_eval.py, the real-trace campaigns by run_tact_hard_eval.py, and the window measurements by run_g1_window.py and run_g1_deepening.py; the three abstention replays of finding (e) by run_abstention_identifiability.py; the planted-channel operating characteristic of finding (f) by run_planted_sensitivity.py. Each script writes the artifact its table cites. Figures 1–4 are vector output from experiments/make_paper_figures.py, which reads the same JSON.

Appendix

TABLE VII
Symbols, in order of first use. Quantities carrying a subscript q are per item; the estimator ships one global scalar unless a group covariate is available (Section VI).

Symbol	Meaning
q, Q	item index; number of items
m_q, K	traces available for item q; voting budget
$a_{q,i}$, a_q^*	answer of trace i; the correct answer
$c_{q,i}$	self-reported confidence, in $(0, 1)$
$y_{q,i}$	$\mathbf{1}[a_{q,i} = a_q^*]$; unobservable at test time
$\bar{p}$	base rate of correct traces
$R_{q,i}$	within-item midrank of $c_{q,i}$ (ties averaged)
$\varphi_{q,i}$	standardized van der Waerden score of $R_{q,i}$; the vote feature
σ_q	realized within-item SD of the normal scores
γ, $\gamma_{\max}$	vote exponent; its cap (4 dev, 2 label-free)
$\hat{a}_q$	the returned answer
D_q	Somers' D within item q, $= 2\,\mathrm{AUC}_q - 1 = 2\,\mathrm{WQD}_q - 1$
N_q	van Elteren pooling weight, $n_q^1 n_q^0$
$\widehat{D}$	pooled signed discrimination
SE_0, SE_J	exact tie-corrected null SE; delete-one-item jackknife SE
SE	$\max\{\mathrm{SE}_0, \mathrm{SE}_J, 1/(2\sqrt{N})\}$
ζ	$\widehat{D}/\mathrm{SE}$; the dead zone is $\lvert\zeta\rvert \le \nu$
ν	significance floor (1.28 dev, 2.33 label-free)
$\tilde{D}$	shrunk discrimination, $\widehat{D}(1 - \nu^2/\zeta^2)_+$
z	$\Phi^{-1}\big((1 + \tilde{D})/2\big)$
M_q	dedup-weighted plurality; the pseudo-label
$g_{q,i}$	$\mathbf{1}[a_{q,i} = M_q]$
$\bar{\rho}$	pair-weighted probability the plurality is wrong
$\hat{\eta}$	estimated attenuation $1 - 2\bar{\rho}$ (not $\bar{p}$)
α, k	split-half agreement; effective number of wrong alternatives
A_{dup}, A_{mgn}	duplicate-collapse and margin-decoupling alarms
A_{root}, A_{size}	split-root ambiguity and gated-item supply alarms
ψ	statistic used by A_{mgn}
κ_c	true confidence–correctness coupling in the oracle (1)
λ_q	per-item coupling under i.i.d. heterogeneity

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